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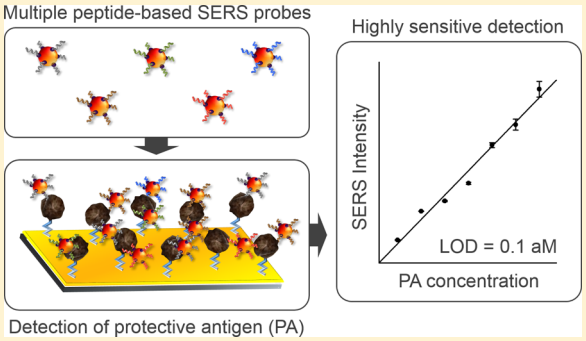
Use of Multiple Peptide-Based SERS Probes Binding to Different Epitopes on a Protein Biomarker To Improve Detection Sensitivity

Kayeong Shin, Jun-Haeng Cho, Moon-Young Yoon, and Hoeil Chung*

Department of Chemistry and Research Institute for Natural Sciences, College of Natural Sciences, Hanyang University, Seoul, 133-791, Korea

ABSTRACT: We propose an analytical strategy to improve the sensitivity for detecting a protein biomarker through signal multiplication by manipulating multiple peptide-based surface-enhanced Raman scattering (SERS) probes to bind the biomarker. Protective antigen (PA) was used as an Anthrax biomarker in this study. For this purpose, five small peptides selective to various PA epitopes with different binding affinities were chosen and peptide-conjugated Au nanoparticle (AuNP) SERS probes were individually prepared using each peptide. Initially, five different SERS probes were separately used to detect PA and the sensitivities were compared. Next, the possibility of enhancing sensitivity by employing multiple SERS probes was examined. Rather than applying the probes simultaneously, which would induce competitive binding, each probe was added sequentially and an optimal probe-addition sequence was determined to provide maximal sensitivity. Finally, PA samples at seven different concentrations were measured with the optimal sequence. The limit of detection (LOD) was 0.1 aM, and the enhancement was more effective at lower PA concentrations. The proposed scheme can be further applicable to detect other protein biomarkers to diagnose various diseases.



Diverse schemes to diagnose various diseases by detecting relevant protein biomarkers have been extensively studied.[1−3] To make the measurements selective, antibodies, aptamers, and peptides are usually incorporated in a probe to specifically recognize target protein biomarkers.[4−8] To make the measurements sensitive, various signal tags, such as fluorescence, electrochemistry and surface-enhanced Raman scattering (SERS) reporters, have been employed.[1−3] Especially, SERS is considerably attractive due to the potential of sensitivity enhancement through enormous increase of Raman intensity by resonance with generated localized surface plasmon on a noble metal surface.[9] Therefore, extensive research has been focused on developing SERS-efficient nanostructures such as particles, rods, and cubes.[10−12]

In parallel with developing SERS-efficient nanostructures, a cooperative approach to improve sensitivity is use of multiple probes able to bind to different epitopes of a target protein. A sandwich immunoassay, also known as a two-site immunoassay, employs two antibodies recognizing different binding sites of a target.[13] Ni et al. used an SERS-based sandwich immunoassay to detect IgG proteins, which improved the sensitivity with a detection limit of about 30 ng/mL (0.2 nM) for measuring rabbit IgG.[14] In addition, the hot spots between Au nanoparticle (AuNP) probes and Au substrate could further increase the overall Raman intensity. Keating et al. reported the SERS spectra of cytochrome c (*Cc*) from several types of sandwich nanoparticle structures (Ag-*Cc*-Au, Au-*Cc*-Au, Ag-*Cc*-Ag).[15] In an Ag-*Cc*-Ag structure, the SERS intensity was four times higher than that of *Cc*-Ag.

Since several epitopes are available in a given protein biomarker, conducting multiple probes to bind to them is potentially advantageous for further sensitivity enhancement compared to that of a sandwich immunoassay; however, it has been rarely exploited. To pursue the goal, a peptide could be advantageous since a peptide-conjugated probe is much smaller than an antibody-conjugated probe.[16−18] The smaller probes reduce the chance of physical congestion when multiple probes bind to a protein, probably yielding a higher binding efficiency. In addition, peptides are chemically and physically stable, so they can withstand denaturation under ambient conditions such as pH, temperature, and ionic strength.[16−18] Also, short peptides with different base sequences are easy to obtain.

Therefore, we attempted to improve the sensitivity for detecting a well-known Anthrax biomarker, protective antigen (PA), through signal multiplication by using multiple peptide-based SERS probes. PA is the central component of the three-part protein toxin secreted by *Bacillus anthracis*, the organism responsible for Anthrax.[19−21] Here, five small peptides (comprising 16 amino acids) selective for various PA epitopes with different binding affinities were selected, as described in Table 1.

Peptide-conjugated AuNP SERS probes were prepared using each peptide. Since the binding affinities of the peptides and their binding regions on PA differed, we initially examined

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Table 1. Amino Acid Sequences and K_d Values[a] of Peptides Used in This Study and Their Binding Regions on PA

peptide	sequence	K_d value (nM)	binding region
P1	HKHAHNYRLPASGGKK	28.0 ± 3.6	PA63
P2	NAYKHHHPPVFYGKGK	4050.8 ± 18.6	PA20
P3	LMPTPHHRLFPMGCGK	626.2 ± 0.6	PA63
P4	LSHNQTIQQDSDGCGK	6855.2 ± 4.9	PA20
P5	TPYYWHHHHIPPGCGK	1285.1 ± 10.5	PA63

[a]Determined by a nonlinear fitting performed on the curve of PA concentration vs fluorescence intensity at 520 nm. Each peptide was tagged with fluorescein isothiocyanate (FITC) on the terminal lysine residue and allowed to bind preincubated PA (0.5 μg/well) for the determination.[16]

Raman intensities in measurement of a 61 nM PA sample using each individual probe. When multiple probes are employed to detect PA, adding each probe sequentially is preferred since it prevents competition for PA binding among probes, which could occur with simultaneous probe addition. Thus, we determined the optimal sequence to add probes to maximize Raman intensity and measured the intensities of seven different PA concentrations from 610 fM to 0.61 aM with the optimal sequence. Then, the variation of the intensities according to the change of PA concentrations was examined. We discussed potential to improve sensitivity by using multiple probes and compared our previous results using a single probe.[22] Meanwhile, dual SERS probes were also allowed to bind on PA together and altered asymmetrical flow field-flow fractionation (AF4) retention due to the increase of PA size by binding with the probes was used as a means for PA detection in our group.[23] The band shift was noticeable in the measurement of 84.3 pM of PA sample; however, the achieved sensitivity was not sufficient for practical applications.

■ EXPERIMENTAL SECTION

Description of the Overall Experimental Procedure. Figure 1 shows a schematic description of the overall

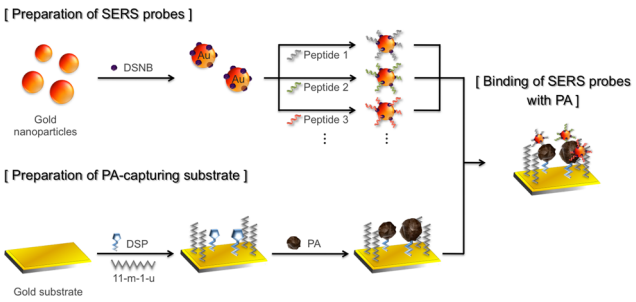

Figure 1. Schematic illustration describing the overall procedure to detect PA with multiple peptide-conjugated SERS probes.

experimental procedure including preparation of peptide-conjugated AuNPs (hereafter, simply referred to as SERS probes), initial PA binding to an Au substrate and application of SERS probes to detect PA. The experimental procedure is mostly analogous to that described in a previous publication.[22] Briefly, dithiobis-*N*-succinimidyl propionate (DSP), which can capture PA, was immobilized on a clean Au substrate. The uncovered surface by DSP was filled with 11-mercapto-1-undecanol (11-m-1-u) to minimize the nonspecific binding of PA or SERS probes on the Au surface. PA is randomly oriented on the surface since PA binding to DSP is not orientation-specific.[24] After the surface passivation, PA samples of seven different concentrations (610 fM, 61 fM, 6.1 fM, 610 aM, 61 aM, 6.1 aM, and 0.61 aM) prepared in a PBS buffer (137 mM NaCl, 2.7 mM KCl, 10 mM Na_2HPO_4, 2 mM KH_2PO_4, pH 7.4) were applied to the substrate. A recombinant PA was used in this study. For purification, the gene encoding PA was subcloned into a pET22b vector (Novagen), expressed in *Escherichia coli*, and purified to homogeneity using a Ni-Sepharose column with denaturing conditions, followed by size-exclusion column chromatography, as previously reported.[16] The remaining active DSP sites were blocked with 2 wt % bovine serum albumin (BSA). Au substrate fully covered with only BSA was used as a control. All chemical reagents used in the study were of the highest available grade, purchased from Sigma-Aldrich and were used as received.

Preparation of SERS Probes and Acquisition of Raman Spectra. Initially, 15 nm AuNPs were synthesized by a well-known citrate reduction method.[25] The average size of the synthesized AuNPs was 16.5 ± 2.1 nm (9.70×10^{12} particles/mL). The 15 nm AuNPs were then grown to approximately 30 and 60 nm AuNPs through a surface-catalyzed reduction of Au^{3+} by NH_2OH, called the seed-mediated growth method.[26,27] The average sizes of the AuNPs were 32.5 ± 2.7 (3.58×10^{11} particles/mL) and 66.5 ± 7.6 nm (2.65×10^{10} particles/mL). The AuNP size distribution was determined by calculating the sizes of dispersed particles in the SEM images (obtained by Hitachi S-4800) using an image analysis program (ImageJ, National Institutes of Health, U.S.A.).

To prepare SERS probes, a Raman reporter molecule (5,5′-dithiobis (succinimidyl-2-nitrobenzoate), DSNB) was conjugated to AuNP surface. The DSNB-labeled AuNPs were centrifuged, and red precipitate was redispersed in 2 mL of a PBS buffer. Five different PA-specific SERS probes were prepared by binding each peptide to DSNB on the surface. The peptides were obtained from Peptron, Korea. The SERS probes were individually allowed to bind PA on the Au substrate for 3 h at room temperature. Uncaptured SERS probes were washed by a series of PBS buffer, 150 mM NaCl solution, DI water, and TBST buffer (20 mM Tris-HCl, 150 mM NaCl, 0.05% Tween-20, pH 8.0).

SERS spectra (resolution: 4 cm^{-1}) were collected by a Raman microscope equipped with a diode laser (λ_{ex} = 785 nm, power = 44 mW), a CCD detector, and a holographic grating (Kaiser Optical Inc., Ann Arbor, MI, U.S.A.). Each sample was positioned on a microscope stage and the laser beam was focused with an objective lens (10×/0.25 NA) to collect spectra. For each sample, 20 SERS spectra (1 s exposure time) were acquired over a 2000 (40 × 50) μm^2 area with a 10 μm interval. The average of the 20 spectra was used for analysis.

■ RESULTS AND DISCUSSION

Investigation of Increased Raman Intensity by Using Multiple SERS Probes. Initially, five different SERS probes (referred to as P1−P5 probes, such as P1 probe designating peptide 1 conjugated AuNP probe) were independently used to measure a 61 nM PA sample. Figure 2 shows Raman spectra of DSNB (a reporter molecule) acquired from the measurement of the PA samples using each SERS probe. The DSNB peaks[28] are clearly apparent in all measurements, and the intensities differ. Since a lower K_d indicated that the peptide had stronger binding affinity for PA, as shown in Table 1, the P1 probe had the strongest binding affinity and the P3 probe had the second strongest affinity.[16,29] The intensities clearly aligned with the

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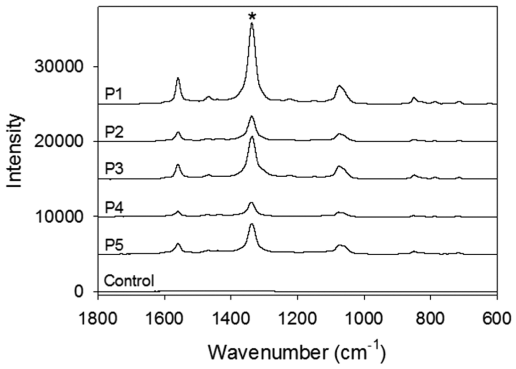

Figure 2. Raman spectra of DSNB (reporter molecule) acquired in the measurement of a 61 nM PA sample using each peptide-conjugated probe (P1−P5) separately. The asterisk indicates the strongest DSNB peak at 1334.5 cm^{-1}.

peptides' K_d values. If a single probe has to be used to detect PA, the P1 probe is the obvious choice, as we found previously.[22] The limit of detection (LOD) for PA using P1 probe was comparable to that of the antibody.[22] As previously confirmed, PA (PA83) is composed of two functional moieties of N-terminal PA20 and C-terminal PA63.[19−21] In a course of toxification, PA 63 initially binds to receptor located at cell surface and PA20 is heptamerized to generate a channel for entry of toxins, EF (edema factor) and LF (lethal factor), into cell. Therefore, the use of SERS probes selective to either PA20 or PA 63 is beneficial to achieve multiple binding on PA. The binding regions of each peptide are presented in Table 1.

To examine the possibility of increasing Raman intensity by employing multiple SERS probes, P1 and P2 probes were initially used to measure a 61 nM PA sample. Figure 3a shows

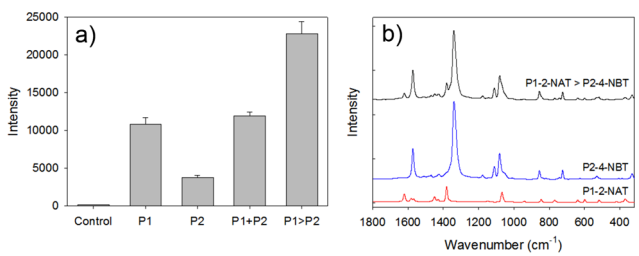

Figure 3. Raman intensities (1334.5 cm^{-1} peak) acquired from a 61 nM PA sample using dual probes (either P1 + P2 or P1 > P2) and each probe separately (P1 or P2) (a). Raman spectra obtained from a 61 nM PA sample using P1−2-NAT (red), P2−4-NBT (blue), and the sequential addition of P1−2-NAT > P2−4-NBT (black) (b). The blue and black spectra are arbitrarily offset for ease comparison of the spectral features.

Raman intensities at 1334.5 cm^{-1} acquired with two probes simultaneously (P1 + P2) and single probe separately (P1 or P2). In comparison with the intensity using the P1 probe alone, the intensity increase with both probes simultaneously (P1 + P2) is minimal. This result indicates that the P1 probe, which has a higher binding affinity, binds to PA dominantly, so P2 binding decreases greatly because it falls behind in the binding competition. Therefore, the use of two probes simultaneously to detect PA is not effective to improve the sensitivity. A better approach to increase Raman intensity could be using probes sequentially to escape competitive binding. For example, a first probe with a high binding affinity is bound to PA, then a second probe with a lower binding affinity is bound to an accessible target epitope. P1 and P2 probes were applied sequentially (designated as P1 > P2) to measure the same sample, and the intensity is shown in the figure. Sequential probe addition increases the intensity by more than 2×. Due to the absence of competitive binding, P2 probe can find its epitope, which increases the overall intensity. To maximize the sensitivity, P1 probes need to use sufficiently to make them bind to all available epitopes. For evaluation, the incubation time of P1 probe in the measurement of a 61 nM PA sample increased from 3 to 18 h, with intervals of 3 h (6 observations), and the resulting intensities at each measurement were examined. There were no significant intensity variations over the tested period, thereby confirming the use of a sufficient amount of P1 probes for analysis.

To alternatively confirm the binding of both probes on PA, P1 and P2 probes were conjugated with different Raman reporters, 2-naphthalenethiol (2-NAT) and 4-nitrobenzenethiol (4-NBT), respectively, and used in the same sequential measurement. To prepare these probes, DSP was used to connect each peptide on the AuNP surface and the reporter molecules were immobilized on the remaining surface through the thiol functional group. Figure 3b shows the spectra acquired from the measurement of a 61 nM PA sample using 2-NAT conjugated to P1 probe (P1−2-NAT, red) and 4-NBT conjugated to P2 probe (P2−4-NBT, blue) separately. The peaks corresponding to 2-NAT and 4-NBT are apparent as previously published.[30,31] The P2−4-NBT intensities are stronger than those of P1−2-NAT, even though P2 has a lower binding affinity. This result is attributed to the greater Raman cross-section of 4-NBT, mainly due to the nitro group. With sequential addition (P1−2-NAT > P2−4-NBT), the spectrum (black) contained features of both the 2-NAT and 4-NBT spectra, indicating that both probes bind to PA. The magnitudes of the 2-NAT peaks are similar for both P1−2-NAT alone and sequential addition of P1−2-NAT > P2−4-NBT. The peak intensities of P2−4-NBT did not vary significantly compared with P2−4-NBT alone.

The enhanced sensitivity for PA detection by using sequential employment of two probes could originate from two sources. PAs that were not bound by the first probe were recognized by the second probe rather than binding of both probes to the same PA simultaneously. Since the PA orientation on the Au substrate is random, a second probe that binds a different epitope increases the chance of detecting PAs. Another possible source of enhanced sensitivity is both probes binding to one PA simultaneously, namely, two-to-one binding. To investigate the occurrence of two-to-one binding, a SEM image of the substrate used to measure the 61 nM PA sample by P1 > P2 was acquired, as shown in Figure 4a. The image shows two-to-one binding as a dimer-like species. Although the two-to-one binding species are occasionally visible, single AuNPs were dominant, as shown in Figure 4b. The observation suggests that one-to-one binding is most common, even with the use of two different probes, and two-to-one binding happens with minor probability.

Observation of Raman Intensity with Two SERS Probes of Different Sizes. With sequential application of probes, using probes of different sizes would influence on the sensitivity since accessibility of the second probe to PA depends on the size of both probes. Two-to-one binding results in much greater hindrance for the second probe in access to PA. To examine the effect of probe size, P1 and P2 probes were prepared with AuNPs of three different sizes (15, 30, and 60

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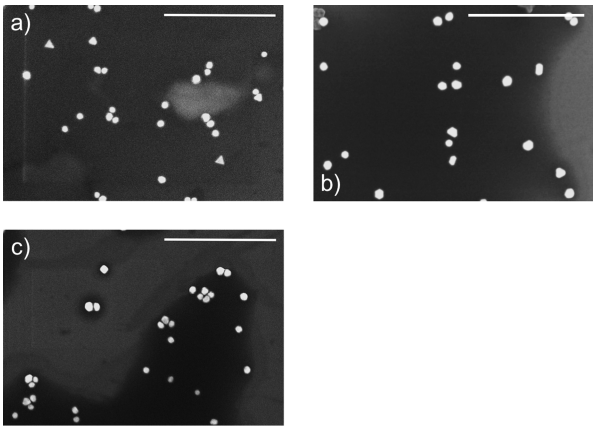

Figure 4. SEM images highlighting two-to-one binding as a dimer-like species (a) and one-to-one binding (b) in the measurement of a 61 nM PA sample with the sequential addition of P1 and P2 probes (P1 > P2). SEM image displaying multimer-like species in measuring the same sample with the sequential addition of all five probes (P1 > P2 > P3 > P5 > P4) is also shown (c). The magnitude of the scale bar is 1 μm.

nm), and DSNB was used as the Raman reporter for all probes. In each case, P1 probe alone and the sequential addition (P1 > P2) were used to measure the 61 nM PA sample. Figure 5

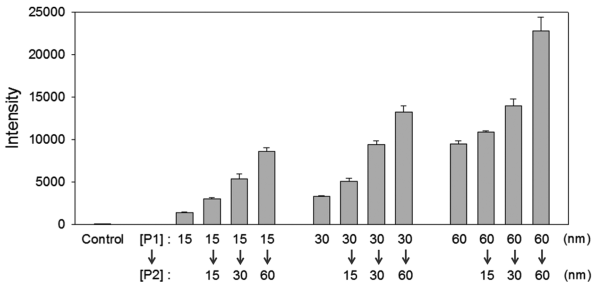

Figure 5. Raman intensities (at 1334.5 cm^{-1}) when P1 and P2 probes of three different sizes are used sequentially to measure a 61 nM PA sample. The arrow indicates the order of probe addition. The intensities acquired using each probe alone are also shown.

shows the Raman intensities of the 1334.5 cm^{-1} peak when the three different sized probes are combinatorially used in the sequential addition. With a single probe, the intensity decreased with the smaller size of the probe, and the 15 nm probe had the lowest intensity. This result was expected since SERS efficiency depends on the AuNP size.[32,33]

When the cases of using P1 (15 nm) and P1 (15 nm) > P2 (15 nm) were compared, the 2.7-fold intensity increase was resultant with the use of dual probes. When the same test was performed using 30 and 60 nm probes, 2.2- and 2.4-fold increase was observed, respectively. There were no meaningful differences in the degrees of signal increase among these three cases. With dual-probes of different sizes, P1 (15 nm) > P2 (60 nm) and P1 (60 nm) > P2 (15 nm), the intensities were similar. Analogous results were obtained with 30/60 and 15/30 nm probe combinations. This result again supports that accessibility of the second probe to PA is not a critical factor governing sensitivity. Therefore, the 60 nm probe provided the highest sensitivity was used for all subsequent experiments.

Determination of Probe-Addition Sequence to Maximize Sensitivity. Since one-to-one binding is most common, the use of additional probes could increase the chance of finding unrecognized PAs and further enhance sensitivity. To determine an optimal probe-addition sequence, the intensities (at 1334.5 cm^{-1}) in the measurement of 61 nM PA sample were examined, using P1 (the high binding affinity) as the first probe and remained probes as the second, as shown in Figure 6.

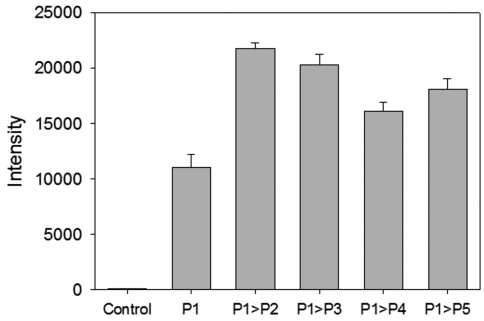

Figure 6. Raman intensities (at 1334.5 cm^{-1}) acquired when P1 probe is initially allowed to bind to PA and a second probe is switched from P2 to P5 for the measurement of a 61 nM PA sample. The intensity acquired using P1 probe alone is also shown.

The intensity of P1 > P3 was slightly lower than that of P1 > P2, although the binding affinity of P3 was stronger than that of P2. Since both P1 and P3 are specific to PA63, as shown in Table 1, P1 is expected to create some degree of physical hindrance for P3 to access the epitope. This hindrance explains the lower intensity for P1 > P3. P1 > P4 had the lowest intensity due to the low P4 binding affinity, even though P1 and P4 bind at different regions. When comparing P1 > P3 and P1 > P5 (P1, P3, and P5 are specific to PA63), P1 > P5 has a lower intensity since P5 has a lower binding affinity compared to P3. Although P1, P3, and P5 are subject to bind to PA63, the size of PA63 is still large and the locations of their epitopes are different. Therefore, the simultaneous binding of these probes is feasible. Overall, the intensities are related to the binding affinities of peptides and their binding region on PA. Based on these observations, the optimal probe-addition sequence was determined to be P1 > P2 > P3 > P5 > P4 to provide the best sensitivity. The intensities from all possible sequences were examined, and P1 > P2 > P3 > P5 > P4 provided the highest intensity, as expected. Figure 4c presents an SEM image acquired from the measurement with five sequential probes. The image shows some multimer-like species, while single particles were dominant in other areas.

Measurement of PA Samples at Different Concentrations by Using Multiple SERS Probes. PA samples at seven different concentrations (610 fM, 61 fM, 6.1 fM, 610 aM, 61 aM, 6.1 aM, and 0.61 aM) were prepared and measured by the probe-addition sequence of P1 > P2 > P3 > P5 > P4. Control measurements (using a substrate covered with BSA only) were also executed at each case. Figure 7 shows the intensities (at 1334.5 cm^{-1}) measured for the samples of various PA concentrations. The intensity obviously increases with increased PA concentration. When the x-axis (PA concentration) was converted to a logarithmic scale, the relationship became linear in the test concentration range, as shown in the inset.

The utility of employing multiple probes to detect PA was evaluated using both 61 nM and 6.1 fM PA samples. Figure 8 shows the intensities (at 1334.5 cm^{-1}) of 61 nM (gray) and 6.1 fM (white) samples, respectively, measured by adding five

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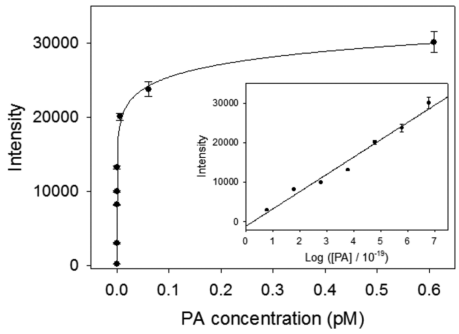

Figure 7. Variation of Raman intensities (at 1334.5 cm^{-1}) in the measurements of PA samples of seven different concentrations (610 fM, 61 fM, 6.1 fM, 610 aM, 61 aM, 6.1 aM, and 0.61 aM) with P1 > P2 > P3 > P5 > P4. The inset shows the same intensities on a logarithmic *x*-axis (PA concentration).

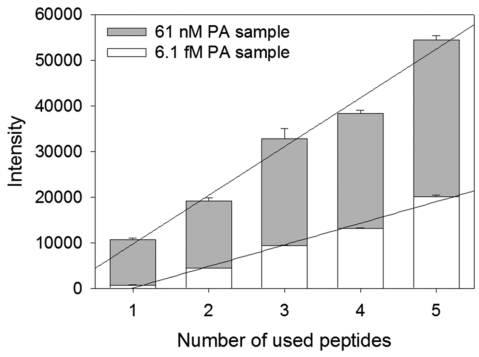

Figure 8. Intensities (at 1334.5 cm^{-1}) acquired with five probes (P1 > P2 > P3 > P5 > P4) added sequentially to measure 61 nM (gray) and 6.1 fM (white) PA samples.

probes sequentially with P1 > P2 > P3 > P5 > P4. With the 61 nM sample, the intensity increased with each additional probe, and the use of all five probes resulted in a 5.1-fold increase compared to P1 alone. With the 6.1 fM sample, the intensity increase was relatively greater, resulting in a 26.5-fold increase. Since one-to-one binding predominates and the increased intensity largely originates from the recognition of PAs that were not bound by previous probes, so the proposed method more effectively enhances sensitivity for measuring samples in low PA concentrations. As a result, the calculated LOD was 0.1 aM.

■ CONCLUSIONS

The use of multiple peptide-based SERS probes improved the sensitivity for PA detection. The sequential addition of individual probes was advantageous for the sensitivity enhancement and more effective when a PA sample in a lower concentration range was measured. Although multiple probes that bind to different PA epitopes were used, one-to-one binding (binding of one probe to one PA) was most common. While the increased occurrence for binding of multiple probes to a single PA, such as two-to-one binding, is helpful for the additional sensitivity improvement.

We are under expanding the proposed method for detection of PA in clinical samples such as serum or plasma. For this purpose, a PA-specific peptide will be conjugated on Au substrate for selective capture of PA in a complex media as previously performed in our previous study[34] and then more SERS-efficient probes will be employed. In addition, optimal linking of several probes into one polyvalent probe is also under investigation to make the analysis faster by avoiding the sequential addition of probes.

■ AUTHOR INFORMATION

Corresponding Author

*E-mail: hoeil@hanyang.ac.kr. Tel.: 82-2-2220-0937. Fax: 82-2-2299-0762.

Notes

The authors declare no competing financial interest.

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